Manuela Dalsoto

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Experience

Altus Sistemas de Automação

Software Development Internship. Jan 2026 - Present

Stacks: Python, C++, C#, .NET, HTML, CSS, JavaScript, Git, GitHub, Bugzilla, Visual Studio, Visual Studio Code

- *Resolved 15–20 software bugs per month by investigating root causes, implementing fixes in C#/.NET, and validating changes against functional and technical requirements.*
- *Developed and maintained automation software for international clients, implementing business rules and customer-specific requirements while collaborating with validation and development teams.*
- *Supported the development of a custom automation platform for Beijer, a Swiss client, adapting software functionality to project-specific requirements and contributing to the delivery of customer-facing automation solutions.*
- *Reduced software regressions by developing and executing functional and integration tests, identifying inconsistencies introduced by changes and validating fixes before delivery.*
- *Integrated and validated physical hardware modules by testing I/O interfaces and communication protocols, connecting technical documentation with practical hardware behavior to identify and troubleshoot integration issues.*
- *Investigated software and hardware issues across the automation stack, collaborating with engineers to identify root causes and technically viable solutions rather than addressing only the reported symptoms.*
- *Contributed to multiple customer projects by balancing software development, debugging, testing, and hardware validation, meeting established delivery deadlines and supporting the broader engineering team.*

Projects

Babel: Communication without barriers

Electronics Capstone Project. Jul 2025 - Nov 2025

- *Built an offline automatic speech translation prototype using a Raspberry Pi 3B, microphone, and speaker, enabling voice translation without internet connectivity.*
- *Developed the translation pipeline in Python, integrating speech recognition, language processing, and text-to-speech components into an end-to-end workflow.*
- *Designed the system to process spoken input locally, reducing reliance on cloud services and enabling operation in environments with limited or no network access.*
- *Implemented audio input and output interfaces to support real-time human-computer interaction through voice commands and translated speech.*
- *Optimized the prototype for resource-constrained hardware, balancing processing requirements, response time, and offline execution on the Raspberry Pi 3B.*
- *Demonstrated the feasibility of an edge-based translation solution for communication in connectivity-constrained environments.*

Educations

Universidade do Vale do Rio dos Sinos

Systems Analysis and Development. Jan 2026 - Present

Software Engineering and Design Patterns, Data Structures and Algorithms, Relational Databases, AI, Machine Learning

Fundação Escola Técnica Liberato Salzano Vieira da Cunha

High School and Electronics Technical Course Jan 2022 - Jul 2026

Digital/Analog Electronics, Microcontrollers, Communication Protocols, Hardware-Software Integration